

Zhanwei Zhang (张展玮)

(+86) 13380806800 | zzhang364@connect.hkust-gz.edu.cn | <https://it-bill.github.io/>

Education

M.Phil. in Data Science and Analytics, Hong Kong University of Science and Technology (Guangzhou) Sep 2025 ~ Present
Information Hub

Advisor: Prof. Zishuo Ding

B.Sc. in Computer Science and Technology, Southern University of Science and Technology (SUSTech) Sep 2021 ~ Jun 2025
Turing Master Class

Advisor: Prof. Yepang Liu

GPA: 3.79 / 4.0 | Weight Avg Score: 90.92 | Ranking: 36 / 195

Main courses: Mathematical Logic (A+), Java Programming (A+), Data Structures & Algorithms (A), Machine Learning (A), Database Systems (A-), Compilers (B+)

Internships

Agent Intern in Qualcomm, Shenzhen May 2026 ~ Present
Core developer of an internal AI Agent platform, evolving its architecture and enabling unified Agent / Workflow / Tool integration.

Designed Engine + Plugin architecture with dynamic backend/frontend loading via Python Entry Points and Module Federation.

Built Agent Runtime on DeepAgents, unifying Tool / MCP / Skills / Subagent integration and supporting AG-UI and HITL.

LLM Intern in Lingsome, Shenzhen Aug 2024 ~ Aug 2025
Designed multi-granularity RAG indexing and joint retrieval strategies to improve domain-data retrieval.

Researched and adapted Microsoft GraphRAG to company-specific domain data, constructing domain knowledge graphs.

Explored automated RAG evaluation, building prototypes for question generation and Ground Truth annotation.

Visiting Researcher in Wuhan University, Wuhan May ~ Aug 2024
Advisor: Prof. Jinfu Chen (WHU); Prof. Weiyi Shang (UWaterloo)

Focused on software logging and failure workarounds.

Developed an automated analysis pipeline to extract, filter, and sample code commits containing try-catch blocks.

Selected Projects

OpenStar Jul 2026 ~ Present
Built Chat / Explore / Watch Agent workflows for open-source project discovery, understanding, and continuous tracking.

Designed Agent research and Watch pipelines to investigate Release / Activity via tools and produce structured assessments.

NIYO.AI Custom-Fit Nail 3D Modeling Platform Jan 2026 ~ Present
Turned vision and geometry prototypes into a production 3D annotation/modeling platform, covering full-stack development and the vision-to-3D workflow.

Built synthetic-data and training pipelines for millions of multi-view samples, model training/evaluation, and production inference.

Rewrote the shared geometry core in Rust with Python/WASM bindings, cutting compute from hundreds of ms to ~10 ms for real-time 3D interaction.

WiTH - AI Health Companion Jun 2026
Built an AI health Agent integrating user profiles, health records, and 20+ Tool Calling capabilities for personalized health interactions.

Implemented semantic memory and proactive health reminders, turning conversations into actionable tasks; won Third Prize at the Shenzhen Hackathon.

AI Micro-Drama Studio Sep 2025 ~ Jan 2026
Built an end-to-end pipeline that transforms novels or scripts into production-ready storyboards.

Unified creation, generation, editing, and export in one interface to streamline publish-ready video production.

Othello Game through Java and Python Programming with Strong AI**Oct ~ Dec 2021 & Mar 2023**

Developed visually appealing interface and implemented Monte Carlo & Alpha Beta Pruning algorithm.

Canteen Traffic Monitoring**Dec 2023 ~ Jan 2024**

Calculated the length of the queue by monitoring data and displayed a chart showing the changes in queue length.

Won award for finalist in National College Students' Innovation and Entrepreneurship Training program.

About 30,000 visits within three months.

Research**Numerical Error Detection in Floating-Point Computing****Sep 2024 ~ Feb 2026**PI-detector: A condition-number-guided perturbation approach can replace costly high-precision oracles, finding 173/174 significant-error cases at about 0.13% of oracle cost (up to $73.46\times$ faster).MGDE: Turn detection into a Newton-Raphson-guided convergent input search, reaching 80 bugs / 47 functions (vs. 70/46 ATOMU, 53/42 FPCC) while being $41.71\times$ and $10.17\times$ faster, respectively.**Reproduction and Evaluation of R1-style Reasoning Pipeline****May 2025 ~ Jun 2025**

Replicated the DeepSeek-R1 training pipeline by implementing GRPO and Cold-start SFT, significantly enhancing the multi-step reasoning capabilities of Qwen2.5 models (0.5B to 7B).

LLM-Based JSON Parser Fuzzing for Bug Discovery and Behavioral Analysis**Sep 2023 ~ Jan 2024**

Used opensource LLMs such as Llama2-7B/13B to generate test cases.

13 JSON Parsers and over 100 types of cases have been tested. Over 26 behavioral diversities have been found.

PublicationsTan, Y., **Zhang, Z.**, Ding, Z., Chen, J., & Shang, W. (2026). *ICE: Reducing search space for error-inducing input detection*. *IEEE Transactions on Software Engineering*, 1-14. [doi:10.1109/TSE.2026.3703362](https://doi.org/10.1109/TSE.2026.3703362)Tan, Y., **Zhang, Z.**, Zhang, H., Zheng, L., Ding, Z., Chen, J., & Shang, W. (2026). *Mathematically-guided detection of floating-point errors*. *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. [ResearchGate](https://arxiv.org/abs/2507.08467) (Accepted)Tan, Y., **Zhang, Z.**, Chen, J., Ding, Z., Xuan, J., & Shang, W. (2025). *Computing floating-point errors by injecting perturbations*. arXiv. <https://arxiv.org/abs/2507.08467>Han, Y., Shen, H., He, X., Mai, Z., Zhang, R., Zheng, Z., Liu, Y., Zhang, X., Li, G., **Zhang, Z.**, Liang, Z., Chen, Y., Xie, Y., Li, M., Shen, G., Wang, C., Ye, J., Zhu, L., Fu, T.-M., & Yang, X. (2025). A comprehensive analysis of interflight variability in carbon dioxide emissions from global aviation. *Environmental Science & Technology*, 59(12), 6179-6191. doi:10.1021/acs.est.5c02371**Patents****一种点餐方法、系统、终端及介质 (Innovative Ordering Method, System, Terminal, and Medium Patent)****May 2023**

Innovated a method and system to alleviate peak-hour traffic in cafeterias.

Applied on May 5, 2023; Application no: 202310498065

Skills**Languages:** English (Fluent; IELTS: 6.5), Mandarin (Native), Cantonese (Native)**Programming Languages & Frameworks:** Java, Python, C/C++, SQL, Spring Boot, Vue, React**Tools:** IntelliJ IDEA, PyCharm, Visual Studio Code, Anaconda, Git, CMake**Honors & Scholarships**

Third Prize, Shenzhen Hackathon

Jun 2026

Special Innovation Award (Unique Winner) & Second Prize, OpenHarmony Competition Training Camp

Sep 2025

Postgraduate Studentship (PGS), HKUST(GZ)

Sep 2025

Outstanding Student, SUSTech

Jan 2024

Honorable Mention, Mathematical Contest in Modeling

May 2023

Finalist, National College Students' Innovation and Entrepreneurship Training program
Third Prize, China Undergraduate Mathematical Contest in Model

Jun 2023
Sep 2023